

TOPIC NAME: AdaBoost

DAY: _____

TIME: _____

DATE: / /

Day	Outlook	Temp	Humidity	Wind	Play tennis	Sample weight
1	Sunny	hot	high	weak	no	1/14
2	Sunny	hot	high	strong	no	1/14
3	Overcast	hot	high	weak	yes	1/14
4	Rain	mild	high	weak	yes	1/14
5	Rain	cool	normal	weak	yes	1/14
6	Rain	cool	normal	strong	no	1/14
7	Overcast	cool	normal	strong	yes	1/14
8	Sunny	mild	high	weak	no	1/14
10	Sunny	cool	normal	weak	yes	1/14
10	Rain	mild	normal	weak	yes	1/14
11	Sunny	mild	normal	strong	yes	1/14
12	Overcast	mild	high	strong	yes	1/14
13	Overcast	hot	normal	weak	yes	1/14
14	Rain	mild	high	strong	no	1/14

Initial Sample weight = $1/N$; N = number of sampleOutlook

Sunny (5): yes = 2, no = 3

Overcast (4): yes = 4, no = 0

Rain (5): yes = 3, no = 2

$$\begin{aligned}
 \text{Gini}(\text{outlook}) &= \frac{5}{14} \left(1 - \left(\frac{2}{5} \right)^2 - \left(\frac{3}{5} \right)^2 \right) + \frac{4}{14} \left(1 - \left(\frac{4}{4} \right)^2 \right) \\
 &\quad + \frac{5}{14} \left(1 - \left(\frac{3}{5} \right)^2 - \left(\frac{2}{5} \right)^2 \right) = 0.343
 \end{aligned}$$

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Temperature

Hot (4) : yes = 2, no = 2

mild (6) : yes = 4, no = 2

cool (4) : yes = 3, no = 1

$$\text{Gini(temp)} = \frac{4}{16} \left(1 - \left(\frac{2}{4} \right)^2 - \left(\frac{2}{4} \right)^2 \right) + \frac{6}{16} \left(1 - \left(\frac{4}{6} \right)^2 - \left(\frac{2}{6} \right)^2 \right)$$

$$+ \frac{4}{16} \left(1 - \left(\frac{3}{4} \right)^2 - \left(\frac{1}{4} \right)^2 \right) = 0.441$$

Humidity

High (7) : yes = 3, no = 4

normal (7) : yes = 6, no = 1

$$\text{Gini(humidity)} = \frac{7}{14} \left(1 - \left(\frac{3}{7} \right)^2 - \left(\frac{4}{7} \right)^2 \right) + \frac{7}{14} \left(1 - \left(\frac{6}{7} \right)^2 - \left(\frac{1}{7} \right)^2 \right)$$

$$= 0.367$$

Wind

strong (6) : yes = 3, no = 3

weak (8) : yes = 6, no = 2

$$\text{Gini(wind)} = \frac{6}{14} \left(1 - \left(\frac{3}{6} \right)^2 - \left(\frac{3}{6} \right)^2 \right) + \frac{8}{14} \left(1 - \left(\frac{6}{8} \right)^2 - \left(\frac{2}{8} \right)^2 \right)$$

$$= 0.429$$

Gini(outlook) is the lowest. So, outlook is the first stump.

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Outlook

sunny

Rain

overcast

sunny (5): yes=2, no=3

overcast (4): yes=4, no=0

Rain (5): yes=3, no=2

$$\text{Total error} = 2(1/14) + 0(1/14) + 2(1/14) = 2/7 = 0.29$$

$$\text{amount of say, } \alpha = \frac{1}{2} \ln \left(\frac{1 - \text{total error}}{\text{total error}} \right) \\ = 0.45$$

New sample weight

For incorrectly classified sample:

$$\text{new sample weight} = \text{sample weight} \times e^{\alpha}$$

$$= 1/14 \times e^{0.45} = 0.11$$

For correctly classified sample:

$$\text{new sample weight} = \text{sample weight} \times e^{-\alpha}$$

$$= 1/14 \times e^{-0.45}$$

$$= 0.046$$

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Day	Outlook	Play tennis	Sample weight	normalized	cumulative normalized
1	Sunny	no	0.046	0.051	0.051
2	Sunny	no	0.046	0.051	0.102
3	overcast	yes	0.046	0.051	0.153
4	rain	yes	0.046	0.051	0.204
5	rain	yes	0.046	0.051	0.255
6	rain	no	0.11	0.122	0.377
7	overcast	yes	0.046	0.051	0.428
8	Sunny	no	0.046	0.051	0.479
9	Sunny	yes	0.11	0.122	0.601
10	rain	yes	0.046	0.051	0.652
11	Sunny	yes	0.11	0.122	0.774
12	overcast	yes	0.046	0.051	0.825
13	overcast	yes	0.046	0.051	0.876
14	rain	no	0.11	0.122	0.998

Normalized sample weight = $\frac{\text{Sample weight}}{\text{summation of all sample weight}}$

GOOD LUCK

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Updated sample weight

Day	Outlook	cumulative normalized	Generated random num
1	Sunny	0.051	0.040
2	Sunny	0.102	0.100
3	overcast	0.153	0.151
4	rain	0.204	0.200
5	rain	0.255	0.250
6	rain	0.377	0.267
7	sunny OC	0.428	0.500
8	sunny	0.479	0.700
9	rain Sunny	0.601	0.990
10	rain	0.652	0.370
11	Sunny	0.774	0.600
12	Sunny OC	0.825	0.682
13	rain OC	0.876	0.886
14	rain	0.998	0.980

Generated random number: Any random number

between 0 to 1.

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New dataset creation with sample weight

Day	Outlooks	Temp	Humidity	wind	Play	sample weight
1	Sunny	hot	high	weak	No	1/14
2	Sunny	hot	high	strong	No	1/14
3	OC	hot	high	W	yes	1/14
4	rain	mild	high	W	yes	1/14
5	rain	cool	normal	W	yes	1/14
6	rain Sunny	cool	normal	S	no	1/14
9	Sunny	cool	normal	W	yes	1/14
11	Sunny rain	mild	normal	S	yes	1/14
14	rain	mild	high	S	no	1/14
6	rain Sunny	cool	normal	S	no	1/14
9	Sunny	cool	normal	W	yes	1/14
11	Sunny	mild	normal	S	yes	1/14
14	rain	mild	high	S	no	1/14
14	rain	mild	high	S	no	1/14

In new dataset, we select day based on generated random number. For instance, in the previous table, the 7th generated random number (0.500) is situated between cumulative (0.479) and 0.601, so here we select the upper range (0.601) and its day, then we include it in new dataset.

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Outlook

Sunny (6) : yes = 4, no = 2

overcast (4) : yes = 1, no = 0

Rain (7) : yes = 2, no = 5

$$\text{Gini (outlook)} = \frac{6}{14} \left(1 - \left(\frac{4}{6} \right)^2 - \left(\frac{2}{6} \right)^2 \right) + \frac{1}{14} \left(1 - \left(\frac{1}{1} \right)^2 \right) + \frac{7}{14} \left(1 - \left(\frac{2}{7} \right)^2 - \left(\frac{5}{7} \right)^2 \right) = 0.4$$

Temperature

Hot (3) : yes = 1, no = 2

Mild (6) : yes = 3, no = 3

Cool (5) : yes = 3, no = 2

$$\text{Gini (temp)} = \frac{3}{14} \left(1 - \left(\frac{1}{3} \right)^2 - \left(\frac{2}{3} \right)^2 \right) + \frac{6}{14} \left(1 - \left(\frac{3}{6} \right)^2 - \left(\frac{3}{6} \right)^2 \right) + \frac{5}{14} \left(1 - \left(\frac{3}{5} \right)^2 - \left(\frac{2}{5} \right)^2 \right) = 0.48$$

Humidity

High (7) : yes = 2, no = 5

Normal (7) : yes = 5, no = 2

$$\text{Gini (humidity)} = \frac{7}{14} \left(1 - \left(\frac{2}{7} \right)^2 - \left(\frac{5}{7} \right)^2 \right) + \frac{7}{14} \left(1 - \left(\frac{5}{7} \right)^2 - \left(\frac{2}{7} \right)^2 \right) = 0.41$$

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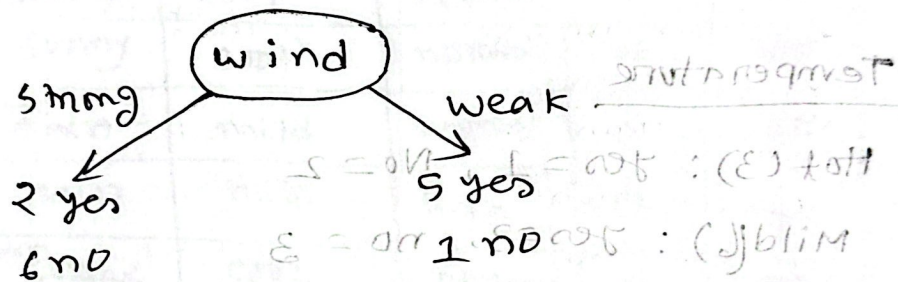
Wind

Strong(8) : yes = 2, no = 6

Weak(6) : yes = 5, no = 1

$$Gini(wind) = \frac{8}{14} \left(1 - \left(\frac{2}{8} \right)^2 - \left(\frac{6}{8} \right)^2 \right) + \frac{6}{14} \left(1 - \left(\frac{5}{6} \right)^2 - \left(\frac{1}{6} \right)^2 \right)$$

$$I.D = \left(\left(\frac{2}{8} \right)^2 + \left(\frac{6}{8} \right)^2 \right) + \left(\left(\frac{5}{6} \right)^2 + \left(\frac{1}{6} \right)^2 \right)$$

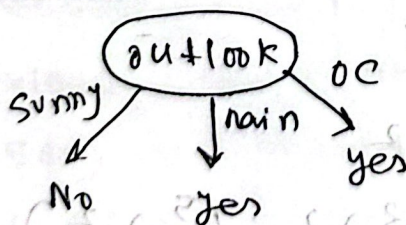


$$\text{Total error} = 2 \left(\frac{1}{14} \right) + 1 \left(\frac{1}{14} \right) = 0.21$$

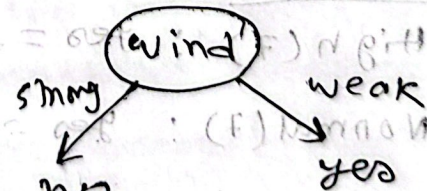
$$\text{Amount of say, } \alpha = \frac{1}{2} \ln \left(\frac{1 - \text{total error}}{\text{total error}} \right)$$

$$\alpha = \frac{1}{2} \ln \left(\frac{1 - 0.21}{0.21} \right) = 0.66$$

Bagging



$$\alpha = 0.45$$



$$\alpha = 0.66$$